

Entropy-Coherence Reciprocity and the Universal Coherence Condition

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Abstract

Formalizes the fundamental reciprocity between entropy and coherence: $dS + k^ d \ln C = 0$. Derives the Universal Coherence Condition (UCC) as a local flow equation governing coherence dynamics. Demonstrates thermodynamic consistency through dissipation bounds and connects information-theoretic and physical projections of the framework. v1.1-c1 (2026-07-08) appends a correction note: the §5.3 Landauer-coherence bound is withdrawn — decided against by the instrumented boundary probe of FRC 100.005 v2.1 — and the closed reciprocity form is read as the locked-limit boundary case of the directional open-system ledger (FRC 566.030 v1.1). The UCC, the*

information-theoretic projections, and the locking-direction free-energy/work relations are unaffected.

1. Introduction

The Second Law of Thermodynamics states that entropy never decreases in isolated systems. Yet coherent structures—crystals, lasers, living organisms—persist and even grow in complexity. How do these observations coexist?

This paper resolves the apparent tension by formalizing a **reciprocity** between entropy S and coherence C : every increase in coherence is paid for by a corresponding increase in entropy elsewhere, and vice versa. This reciprocity is not an approximation or a limit—it is exact, holding at every scale from quantum systems to cosmological horizons.

From this reciprocity, we derive the **Universal Coherence Condition (UCC)**—a local partial differential equation governing the flow of coherence in space and time. The UCC is to coherence what the heat equation is to temperature: a transport equation with sources, sinks, and dissipation.

Entropy-Coherence Reciprocity

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2. Definitions and Units

2.1 Coherence

Coherence C is a dimensionless scalar in the range $(0, 1]$:

$$C = \frac{1}{N} \sum_{i < j} \cos(\phi_i - \phi_j)$$

where ϕ_i are the phases of the constituent oscillators. Perfect phase alignment gives $C = 1$; random phases give $C \rightarrow 0$ ([FRC-100-001]).

2.2 The Coherence Constant

We define the coherence constant k^* with two conventions depending on the domain:

Domain	Value of k^*	Units
Information/cognition	1	nats
Thermodynamic/physical	k_B	J/K

Quantum	$\hbar$	$J \cdot s$
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The choice of k^* determines whether the reciprocity law operates in information-theoretic or thermodynamic units. All results are structurally identical regardless of convention.

2.3 Regularization

For numerical computation, we regularize the coherence to avoid $\ln 0$ divergences:

$$C_{\epsilon} = \frac{1}{Z^2 + \epsilon}, \quad \epsilon > 0 \text{ (small)}$$

where Z is a partition-function-like normalization. In the limit $\epsilon \rightarrow 0$, exact results are recovered.

3. The Reciprocity Law

3.1 Statement

The fundamental reciprocity between entropy and coherence:

$$dS + k^* \, d \ln C = 0 \quad \text{(closed system, reversible)}$$

This integrates to:

$$S + k^* \ln C = \text{const.} \quad \text{(conservation form)}$$

The constant is set by initial conditions. This is the **entropy-coherence conservation law**: the sum of thermodynamic entropy and coherence entropy is conserved in reversible processes. This closed form is the reversible/locked-limit boundary case of the open directional ledger, not the general open-system law (per [FRC-566-030] v1.1).

3.2 Irreversible Generalization

For irreversible processes, the reciprocity becomes an inequality:

$$dS_{\text{total}} + k^* \, d \ln C \geq 0$$

where $S_{\text{total}} = S_{\text{sys}} + S_{\text{env}}$. The irreversible entropy production $\sigma_{\text{irr}} \geq 0$ is always non-negative:

$$\sigma_{\text{irr}} = dS_{\text{total}} + k^* \, d \ln C \geq 0$$

This ensures the Second Law is never violated—coherence can increase locally only at the cost of entropy production elsewhere ([FRC-100-005]). Per the 2026-07-08 correction, this combined inequality is a locking-direction ($d \ln C > 0$) statement: the sign theorem of stochastic

thermodynamics applies to dS_{total} alone, and on quasi-static unlocking the combined form is measurably negative ([FRC-566-030] v1.1).

3.3 Physical Interpretation

The reciprocity law encodes a fundamental trade-off:

Process	dS	$d \ln C$	Interpretation
Decoherence	> 0	< 0	Entropy absorbs lost coherence
Phase-locking	< 0 (local)	> 0	Coherence costs entropy
Thermal equilibrium	Maximum	$-\infty$ ($C \rightarrow 0$)	Ground state
Pure state	Minimum	0 ($C = 1$)	Maximum coherence

Reciprocity Landscape

Reciprocity Landscape

4. Information-Theoretic Projection

4.1 Shannon Entropy and Coherence

For a probability distribution p , define the information-theoretic coherence:

$$C[p] = \exp\left(-\frac{H(p)}{k^*}\right)$$

where $H(p) = -\sum p_i \ln p_i$ is the Shannon entropy in nats. This maps distributions to coherence values:

- Uniform distribution: $H = \ln N$, $C = N^{-1/k^*}$ (minimum coherence)
- Delta distribution: $H = 0$, $C = 1$ (maximum coherence)

4.2 KL Divergence as Coherence Ratio

The Kullback-Leibler divergence between distributions yields a coherence ratio:

$$\mathrm{RER}(p \rightarrow q) = \frac{C[q]}{C[p]} = \exp\left(-\frac{D_{\text{KL}}(p \parallel q)}{k^*}\right)$$

This is the **Relative Entropy-coherence Ratio**: it measures how much coherence is lost (or gained) in transitioning from distribution p to q .

4.3 Mutual Information and Joint Coherence

For joint distributions:

$$I(X; Y) = D_{\text{KL}}(p_{XY} \parallel p_X p_Y)$$

The joint coherence decomposes as:

$$C_{XY} = C_X + C_Y - \exp\left(-\frac{I(X; Y)}{k^*}\right)$$

Mutual information *reduces* joint coherence—correlated variables are less coherent than independent ones. This has deep implications for entanglement ([FRC-100-003]).

5. Thermodynamic Projection

5.1 Free Energy and Coherence

The Gibbs free energy in the FRC framework:

$$G = H_{\text{enthalpy}} - TS - k^* T \ln C$$

The spontaneity condition becomes:

$$\Delta G = -k^* T \ln C < 0$$

Processes that increase coherence decrease free energy—they are thermodynamically spontaneous. This provides the thermodynamic driving force for self-organization ([FRC-100-005]).

5.2 Work Extraction

The maximum work extractable from a coherence difference:

$$W_{\text{max}} = k^* T \ln C \quad \text{(reversible limit)}$$

This connects coherence to the resource theory of thermodynamics: coherence is a thermodynamic resource from which work can be extracted.

5.3 Correction (2026-07-08): Landauer-Coherence Bound Withdrawn

The original §5.3 proposed a Landauer-flavored floor on coherence erasure, $Q_{\text{erasure}} \geq k^* T \ln C$. **This bound is withdrawn.** It was decided against by the instrumented boundary probe of [FRC-100-005] v2.1: on quasi-static unlocking, total entropy production falls through the proposed floor with no plateau (combined form -0.103 ± 0.001 at the slowest ramp, against an asymptote of -0.132), invariant under every $k^* > 0$, with heat flowing from bath into system.

The corrected directional statement: **building coherence always produces entropy** — on locking branches, $dS_{\text{total}} + k^{\{ \} } d \ln C \geq dS_{\text{total}} \geq 0$. **Destroying coherence carries no universal floor** — on quasi-static unlocking, $dS_{\text{total}} \rightarrow 0$ while $d \ln C < 0$, so the combined form is negative and the erasure cost vanishes in the slow limit. The Landauer analogy applies to the locking direction only: bit erasure lowers the system's entropy (a locking operation, where the bound holds); coherence destruction raises* it, so the system pays its own ledger and the bath owes nothing.

Lineage: original §5.3 claim recorded results-negative-kept; original text visible in the v1 record (DOI 10.5281/zenodo.17437759); audit in FRC 828.001. §5.1–5.2 are locking-direction statements and stand.

Thermodynamic Projection

Thermodynamic Projection

6. The Universal Coherence Condition (UCC)

6.1 Derivation

Starting from the reciprocity law in local form, we derive the UCC by requiring coherence to satisfy a conservation law with sources:

$$\partial_t (\ln C) = -\nabla \cdot J_C + S_C$$

where:

- $J_C = -D_C \nabla (\ln C)$ is the **coherence current** (Fick's law for coherence)
- $D_C > 0$ is the **coherence diffusion coefficient**
- S_C is the **coherence source/sink term**

6.2 Expanded Form

Substituting the current into the conservation law:

$$\partial_t (\ln C) = D_C \nabla^2 (\ln C) + D_C \nabla \cdot \nabla (\ln C)^2 + S_C$$

This is a nonlinear diffusion equation. The quadratic gradient term $\nabla \cdot \nabla (\ln C)^2$ is always positive, providing a self-amplification mechanism: regions of steep coherence gradient spontaneously steepen further.

6.3 Steady-State Solutions

At steady state ($\partial_t \ln C = 0$):

$$[D_C \nabla^2 (\ln C) + D_C \nabla \cdot \nabla (\ln C) + S_C = 0]$$

For the source-free case ($S_C = 0$), the substitution $u = \ln C$ yields:

$$[D_C \left(\nabla^2 u + \nabla u \right) = 0]$$

This has solutions $u = \text{const}$ (uniform coherence) or exponential profiles $C(x) \propto e^{-\alpha x}$ (exponential decay from a coherent core).

6.4 Connection to the Lambda-Field

The UCC is the continuity equation for the Lambda-field current ([FRC-100-007]). Using $\Lambda = \Lambda_0 \ln C$:

$$[\partial_t \Lambda = -\Lambda_0 \nabla \cdot J_C + \Lambda_0 S_C]$$

The coherence current becomes:

$$[J_C^\mu = -\frac{D_C}{\Lambda_0} \partial^\mu \Lambda]$$

This is the covariant form of the UCC, valid in relativistic settings.

UCC Flow Diagram

UCC Flow Diagram

7. Dissipation and Stability

7.1 Dissipation Functional

Define the total dissipation rate:

$$[\sigma(t) = k^* D_C \int \nabla^2 \ln C \, dV \geq 0]$$

This functional is always non-negative, ensuring that the UCC dynamics never violate the Second Law. The dissipation rate measures how fast coherence gradients are being smoothed out.

7.2 Lyapunov Functional

The free energy functional:

$$[F[C] = -k^* T \int \ln C \, dV]$$

is a Lyapunov functional for the UCC dynamics:

$$[\frac{dF}{dt} = -\sigma(t) \leq 0]$$

This proves that UCC dynamics are thermodynamically stable: the system always evolves toward lower free energy (higher coherence) until dissipation vanishes at equilibrium.

7.3 Equilibrium Conditions

At equilibrium ($\sigma = 0$):

$$\nabla(\ln C) = 0 \implies C = \text{const.} \quad \text{(uniform coherence)}$$

or in the presence of sources:

$$D_C \nabla^2 (\ln C) = -S_C \quad \text{(Poisson equation for coherence)}$$

The equilibrium coherence profile is determined by the spatial distribution of sources and sinks.

8. Multi-Scale Structure

8.1 The Mu-Level Hierarchy

The UCC operates across multiple scales, organized by the **mu-levels** of the FRC framework:

Mu-Level	Scale	D_C	Physical System
μ_0	Planck	$\sim \hbar/m_P$	Quantum vacuum
μ_1	Atomic	$\sim 10^{-10} \text{ m}^2/\text{s}$	Electrons, photons
μ_2	Molecular	$\sim 10^{-6} \text{ m}^2/\text{s}$	Biomolecules
μ_3	Neural	$\sim 10^{-2} \text{ m}^2/\text{s}$	Brain dynamics
μ_4	Social	$\sim 10^2 \text{ m}^2/\text{s}$	Collective behavior

The diffusion coefficient D_C increases with scale, reflecting faster coherence transport at larger scales.

8.2 Scale Coupling

Adjacent mu-levels are coupled through the source term:

$$S_C^{(n)} = \eta_{n,n-1} (C^{(n-1)} - C^{(n)}) + \eta_{n,n+1} (C^{(n+1)} - C^{(n)})$$

This creates a cascade: coherence at fine scales (μ_0) sources coherence at coarser scales ($\mu_1, \mu_2, \dots$), while decoherence at coarse scales (μ_3) destroys coherence at finer scales.

9. Applications

9.1 Quantum Measurement

During measurement ([FRC-100-003]):

- Local coherence increases rapidly ($C_{\text{sys}} \rightarrow 1$ for the selected eigenstate)
- Environmental coherence decreases ($C_{\text{env}} \rightarrow 0$)
- Total: $dS_{\text{total}} + k^* d \ln C_{\text{total}} \geq 0$ (satisfied)

9.2 Biological Systems

Living systems maintain $C > 0$ against thermal decoherence by continuously producing entropy:

$$\frac{dS_{\text{env}}}{dt} \geq k^* \left| \frac{d \ln C_{\text{bio}}}{dt} \right|$$

This sets a **minimum metabolic rate** for coherence maintenance ([FRC-100-005]).

9.3 Quantum Computing

Quantum error correction can be viewed as maintaining coherence against UCC dissipation:

$$P_{\text{error_correction}} \geq \frac{\sigma(t)}{k^* T}$$

The power required for error correction must exceed the coherence dissipation rate.

10. Experimental Predictions

10.1 Measurable Quantities

The UCC makes predictions for:

- Decoherence rates:** $\gamma_{\text{decoherence}} = D_C k_{\text{min}}^2$ (related to the minimum wavenumber of the system)
- Coherence profiles:** Exponential decay $C(r) \propto e^{-r/\xi_C}$ with coherence length $\xi_C = \sqrt{D_C / \gamma}$
- Free energy changes:** $\Delta G = -k^* T \Delta \ln C$ (measurable via calorimetry)
- Born rule deviations:** Out-of-equilibrium coherence produces $\delta P \in [10^{-4}, 10^{-3}]$ ([FRC-100-006])

10.2 Distinguishing FRC from Standard Decoherence

Observable	Standard Decoherence	UCC Prediction
Decay profile	Exponential in time	Exponential in space and time
Source dependence	Environment-only	Self-sourcing (gradient term)
Equilibrium	$\mathcal{C} = 0$ (thermal)	$\mathcal{C} > 0$ (Lambda-field minimum)
Recoherence	Forbidden	Allowed (coherence echoes)

The most distinctive prediction is **coherence echoes**: partial recovery of coherence after measurement under specific phase-reversal protocols.

11. Conclusion

The entropy-coherence reciprocity ($dS + k^* d \ln C = 0$) and the Universal Coherence Condition provide the thermodynamic backbone of the FRC framework. Together they ensure:

- Thermodynamic consistency:** No perpetual motion, no entropy decrease in isolation
- Dynamical content:** A PDE governing coherence flow in space and time
- Predictive power:** Specific, measurable predictions for decoherence rates, coherence profiles, and Born rule deviations
- Universal applicability:** The same equations operate from quantum to macroscopic scales

The UCC is the master equation of coherence. All other FRC results—collapse ([FRC-100-003]), Born rule ([FRC-100-006]), Lambda-field dynamics ([FRC-100-007])—are special cases or consequences of this fundamental flow equation.

Connections

- [FRC-100-001] — Coherence definition and FRC operator
- [FRC-100-003] — Measurement as coherence flow (collapse dynamics)
- [FRC-100-005] — Thermodynamic consistency (detailed proof)
- [FRC-100-006] — Born rule as UCC equilibrium
- [FRC-100-007] — Lambda-field as covariant UCC variable
- [FRC-566-030] — Directional open-system ledger; reciprocity current computed exactly (v1.1)

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